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Quick Response (QR) codes for patient information delivery: A digital innovation during the coronavirus pandemic

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Abstract

Over the past year, Quick Response (QR) codes have played a significant role in our day-to-day lives in reducing the transmission and tracking the spread of COVID-19. In this article, we share our innovation utilising QR codes to replace paper information leaflets allowing patients to immediately access the required information on their own personal device. This is contactless and therefore preferred to reduce viral transmission, as well as having several other advantages. Our findings demonstrate that QR codes are a familiar, easy-to-use system and a preferred tool for delivering patient information over paper leaflets. The findings and methodology may be of benefit to other units seeking to improve their infection control in the COVID-19 era.

Keywords

Quick Response codes, QR codes, orthodontics, patient information leaflets, COVID-19, pandemic

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Introduction

COVID-19, a contagious disease caused by the novel virus SARS-CoV-2, was announced as a global pandemic on 11 March 2020. According to the World Health Organization (WHO), transmission of the virus can occur either through direct, indirect or close contact with infected personnel (WHO, 2020a). In a dental setting, this can also occur via airborne or fomite (contaminated surfaces) transmission (WHO, 2020b). To tackle the spread of the virus, the Chief Dental Officer issued a letter on 25 March 2020 stating that 'all routine, non-urgent dental treatments should be stopped, including orthodontics' (NHS England, 2020). On 20 August 2020, Public Health England issued a guidance for the remobilisation of services within the health and care settings with infection prevention and control recommendations; one of the key points was safe management of the environment with frequent cleaning of surfaces (Public Health England, 2020). To reinforce infection prevention procedures, we advocated the use of Quick Response (QR) codes in delivering patient information leaflets (PILs), thus minimising the need for cross-contact and limiting the spread of infection.

QR codes are not a new concept; however, recently, with social distancing and the fear of contacting contaminated surfaces, QR codes have played an integral role during the pandemic in many sectors, more specifically the NHS Track and Trace and NHS COVID-19 application. They have also been used widely across many other businesses, particularly restaurants, to facilitate contactless selection and ordering of food. QR codes are two-dimensional barcodes presented in black and white mosaic patterns that can store a surplus of information such as patient details, radiographic interpretation and treatment plans (Shakil et al., 2015). With the use of smartphones, integrated cameras can be used to read these visual codes. In addition, QR code scanners can be easily downloaded on mobile devices and efficiently decode these barcodes (Shakil et al., 2015; Vazquez-Briseno et al., 2012).

QR codes are widely used in healthcare. For example, in the medical field, they are printed as wristbands encoding for a patient's identity (Kubben, 2011), printed on patient's

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test tubes as well as on biopsy reports, thus, safely maintaining records (Shakil et al., 2014). In addition, QR codes offer easy and fast recognition of information sources. They can be incorporated in a brochure, linking the patient to further information on a website (Kubben, 2011). In dentistry, QR codes have a wide range of applications such as dental education, practice management and product promotion (Kafadarova et al., 2019). In the oral and maxillofacial specialty, Shakil et al. (2014) found that these 2D codes can be presented to patients for oral health information as well as linked to short videos describing the dental surgical procedures and providing postoperative instructions. In orthodontics, because of the longitudinal nature of the treatment, QR codes can be an excellent tool in record keeping (Singh and Aggarwal, 2017). More recently, the British Orthodontics Society (BOS) has generated QR codes for all PILs and made them available on their website encouraging members and non-members to use them in their department and practices (BOS, 2021).

However, data that investigated the use of QR codes in orthodontics are limited. To our knowledge, there are limited studies that assessed patient use of QR codes, their preferences or efficiency and effectiveness of the QR codes in providing electronic patient information leaflets. Thus, the aim of the present study was to explore: (1) patient awareness of QR codes and the modality of use; (2) the ease of use of QR codes as a digital tool in delivering patient information; and (3) patient preferences of receiving electronic versus paper PILs.

Materials and methods

This prospective study was conducted at the Queen Victoria Hospital NHS Foundation Trust (QVH) orthodontic department from the period of 1–16 September 2020. Approval was granted from the Quality Control and Governance Team on 27 August 2020. Our sample size was based on a previous study that recruited 79 participants assessing QR code familiarity and usage (Ozkaya et al., 2015). Post hoc statistical power analysis was also conducted assuming a null hypothesis of the proportion of patients using the QR code is 0.5, difference of 0.65, a significance of $\alpha = 0.05$ and a power of 0.9, giving a sample size of 107. The post hoc power analysis was conducted using Stata 16.1 (StataCorp, 2019). Although the use post hoc power analysis is insubstantial (Bacchetti, 2002; Hoening and Heisey, 2001), this has been included so researchers can use it as a basis for future experimental design.

A specific questionnaire was developed that explored patients' awareness of QR codes, methods used to scan the code and the ease of use of this electronic method compared to a paper leaflet and patient's preference (Figure 1). All patients or parents accompanying the patient receiving orthodontic treatment were invited to participate. Demographics were recorded for the individual who scanned the QR code. There were no age restrictions. Participants who did not have their smartphones were excluded.

Creation of QR code

The process of generating a QR code was as follows: (1) three PILs were reformatted into an A4 pdf format to be user friendly on smartphones (Figure 2); (2) the reformatted patient information leaflet was loaded onto the Trust hospital website under the patient information leaflet section of the Orthodontic Department webpage. Using the URL weblink for this specific webpage, QR codes were generated by our photography team using a free web-based application; (3) laminated sheets of these QR codes were then placed in accessible areas in all surgery rooms (Figure 3); (4) patients receiving treatment or the parent accompanying the child were then guided to scan this visual code and read through the linked PIL; (5) after that, the person who scanned the QR code was invited to complete a paper questionnaire (Figure 1), with pens wiped after each use to prevent cross-contamination. Should the QR code not work, or the patient declined participation, paper PILs were provided instead. We recorded data for those where the QR code did not work and those that did not consent to participate. The age and gender of the person scanning the QR code was also recorded. Data were tabulated within a Microsoft Excel database. In addition, descriptive analyses were performed using SPSS version 26.0 (SPSS Inc, Chicago, IL, USA). Using Stata 17 Statistical Software, multinomial logistic regression was used to determine the effect of age group and gender on the response variables (Agresti, 1996: StataCorp, 2019).

Results

A total of 109 consecutive orthodontic respondents completed the questionnaire, and a 100% response rate was achieved. The age range was 10–62 years (mean age = 26.5 years), of which 41% were aged under 18 years. Male-to-female ratio was 0.6:1. Out of 109 participants, 71% had previously used QR codes while 32% had never used them before (Figure 4). Of the cohort, 82% required no assistance to scan QR codes (Figure 4). The inbuilt QR reader on the cameras of the smartphones was most widely used (80%) whereas 16% of the participants used a QR reader application that was downloaded on their smartphone/device (Figure 5). Only 4% reported that scanning the QR code did not work. Although 3% reported the use of QR codes neither easy nor difficult and 3% found the scanning difficult, the majority reported QR codes 'Easy to use' (94%) (Figure 6). When patients were asked about their preferences to access patient information, only 2% reported paper leaflets, 26% were neutral while the majority (72%) preferred accessing an electronic leaflet via a QR code instead of a paper leaflet (Figure 7).

Figure 1. Quick Response code patient questionnaire.

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Quick Response Code Patient Questionnaire

Date --/--/2020

Form to be completed by the user

Do you consent to try the QR Code Yes No

This is an example of a QR code. They are a bit like a barcode in a shop. If scanned by a device (like a smartphone), they give the user access to information. At QVH, we are thinking of using QR codes to give our patients access to patient information leaflets as an alternative to giving out paper leaflets.

Demographics

Please circle the answer

Age	
Male	Female

Questionnaire

Please circle the answer

Questions			
1- Are you familiar with Quick Response (QR) codes?	Yes	No	
2- What did you use to scan the QR code?	Camera	QR Scanner app	Did not work
3- Did you need any assistance?	Yes	No	
4- How did you find the use of QR code compared to a leaflet?	Easier	Neither easy nor difficult	More difficult
5- Would you prefer using QR code again rather than a leaflet?	Yes	No	

It is difficult to determine the impact of previous QR usage on patient ease of use and patient preference for QR or paper leaflet as the non-QR group is small, so any statistical analysis is unreliable.

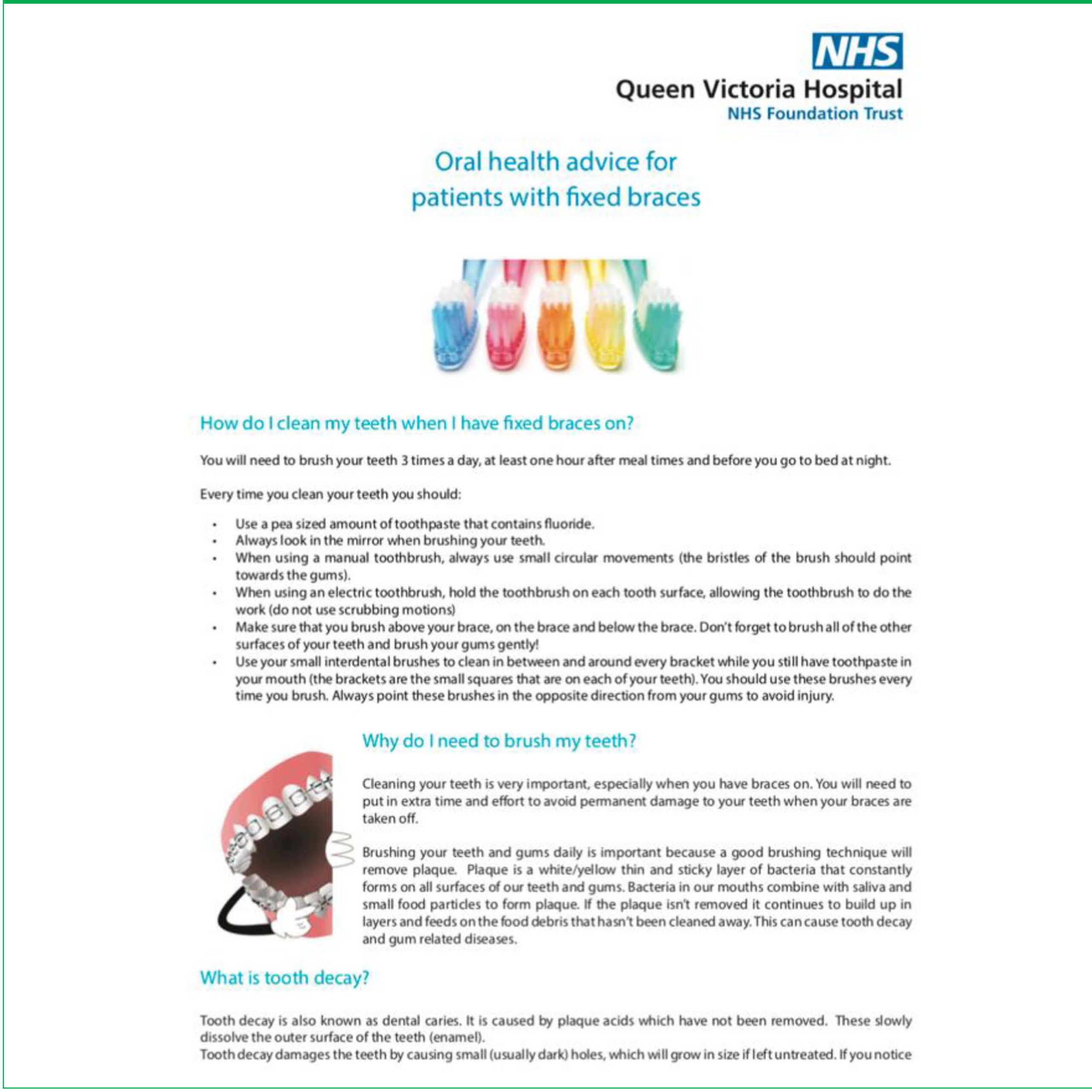
With regard to age, 41% of the participants who completed the survey were aged under 18 years. Of those, 67% preferred the use of an electronic method over a paper leaflet in providing patient information compared to 33% who were neutral. Of those participants above the age of 18 years, 76% reported higher preference to electronic leaflets compared to 20% who had no preference and 4% who preferred paper leaflets. However, patients' preferences or any other variable were not significantly related to age nor gender (Figure 8).

Discussion

Awareness of QR codes

Before the COVID-19 pandemic, an earlier exploratory study reported that the familiarity with QR codes was low among college students (44%) and only 68% used them for seeking information and purchasing, suggesting the novelty of the QR codes attributing to the low awareness levels (Ozkaya et al., 2015). Nevertheless, during the pandemic, the availability of the QR codes increased exponentially with the creation of NHS QR codes for hospitality industries, tourism and leisure centres, libraries and community centres (Department of Health and Social Care, 2020). In addition, there has been wide implementation of this digital

Figure 2. An example of a patient information leaflet in an A4 pdf format.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Oral health advice for patients with fixed braces

How do I clean my teeth when I have fixed braces on?

You will need to brush your teeth 3 times a day, at least one hour after meal times and before you go to bed at night.

Every time you clean your teeth you should:

- Use a pea sized amount of toothpaste that contains fluoride.
- Always look in the mirror when brushing your teeth.
- When using a manual toothbrush, always use small circular movements (the bristles of the brush should point towards the gums).
- When using an electric toothbrush, hold the toothbrush on each tooth surface, allowing the toothbrush to do the work (do not use scrubbing motions)
- Make sure that you brush above your brace, on the brace and below the brace. Don't forget to brush all of the other surfaces of your teeth and brush your gums gently!
- Use your small interdental brushes to clean in between and around every bracket while you still have toothpaste in your mouth (the brackets are the small squares that are on each of your teeth). You should use these brushes every time you brush. Always point these brushes in the opposite direction from your gums to avoid injury.

Why do I need to brush my teeth?

Cleaning your teeth is very important, especially when you have braces on. You will need to put in extra time and effort to avoid permanent damage to your teeth when your braces are taken off.

Brushing your teeth and gums daily is important because a good brushing technique will remove plaque. Plaque is a white/yellow thin and sticky layer of bacteria that constantly forms on all surfaces of our teeth and gums. Bacteria in our mouths combine with saliva and small food particles to form plaque. If the plaque isn't removed it continues to build up in layers and feeds on the food debris that hasn't been cleaned away. This can cause tooth decay and gum related diseases.

What is tooth decay?

Tooth decay is also known as dental caries. It is caused by plaque acids which have not been removed. These slowly dissolve the outer surface of the teeth (enamel).
Tooth decay damages the teeth by causing small (usually dark) holes, which will grow in size if left untreated. If you notice

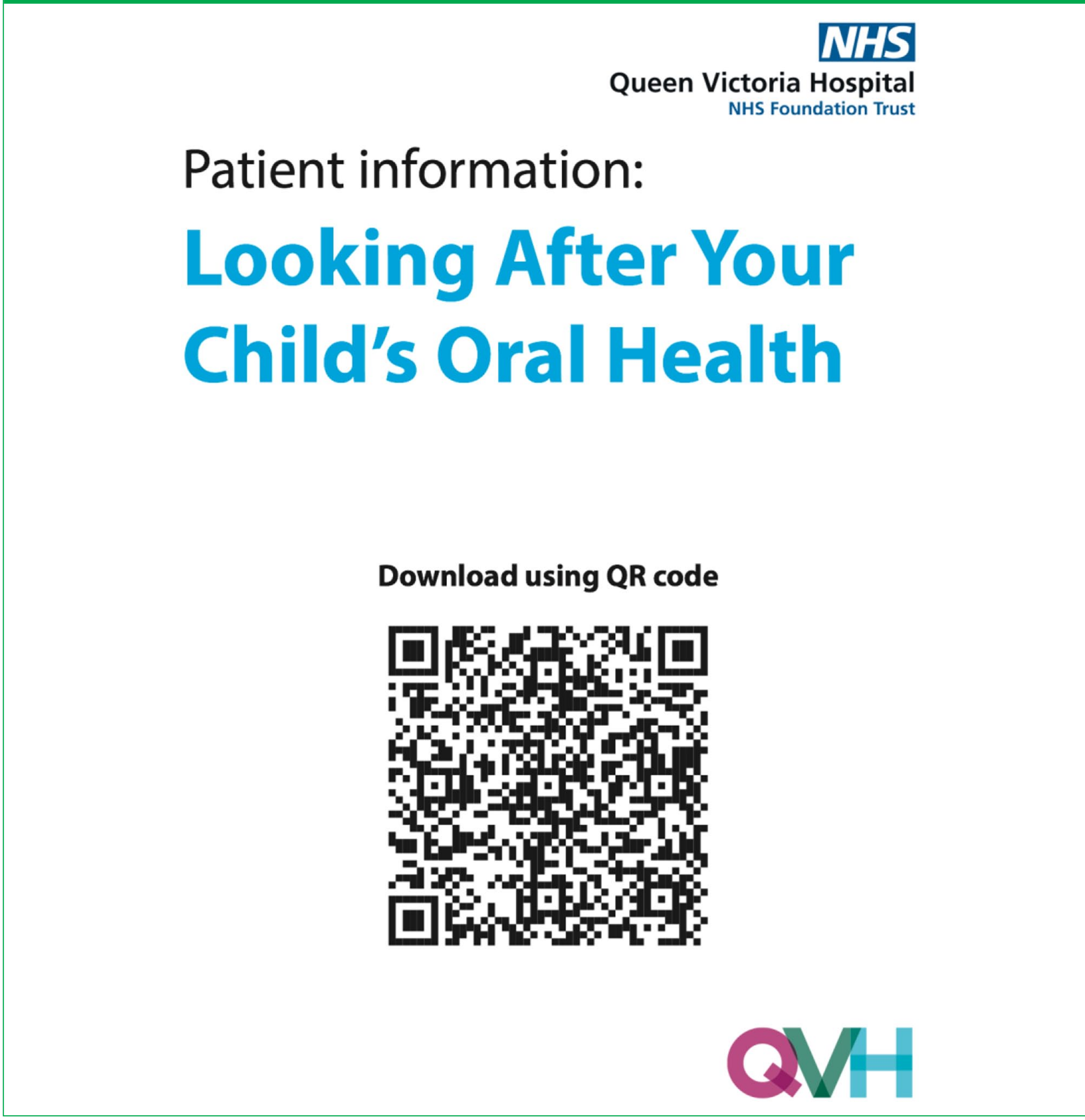
technology into policy and healthcare for pandemic planning, contact tracing and quarantine (Whitelaw et al., 2020). We investigated patient awareness of QR codes and the method used to scan them, and found that the majority of the participants were familiar with QR codes with no assistance required to scan them.

Further research is required to assess the 'efficiency' and 'effectiveness' of the QR code versus patient leaflet in terms of time taken for downloading QR codes and investigating patient understanding and information retention, with an appropriate sample size for inferential statistical analyses. Further studies could also explore the impact of previous QR usage on ease of use of QR codes and patient preferences for QR code or paper leaflets which is beyond the aims of this study.

Applications of QR Codes

QR codes in healthcare systems. In the wake of the global pandemic, QR codes became an essential tool that linked the virtual world to our physical life, minimising friction and increasing engagement. We explored the use of QR codes in delivering patient information leaflets as a digital tool to minimise patient contact with surfaces and thus limit

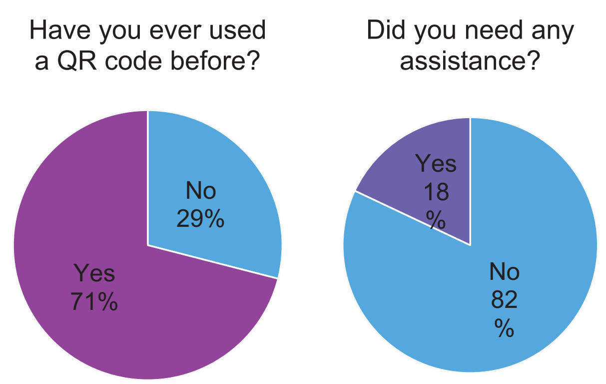
Figure 3. An example of a generated Quick Response code.



cross-infection. The majority of participants reported that scanning the QR code was 'Easy' (94%). The successful set-up process of QR code was a collaborative work between several departments at QVH, which rendered the whole process cost-effective.

Currently, this project has extended its use to the Breast Care Team and Maxillofacial department at QVH. Other applications of QR code in other healthcare systems have also shown to be very promising (Dixon et al., 2013; Uzun and Bilgin, 2016). For example, in 2013, Dixon and colleagues conducted a pilot study investigating the feasibility of the use of QR codes for the surgical safety process. With the use of QR codes, the time-out process was efficient, easy to handle, provided the required information and was successful and accurate (64%, 77%, 89%, 92.5% and 100%, respectively). Moreover, 74% of the surgical team preferred this new method compared to the current time-out procedures, operation site marking and WHO surgical safety checklist (Dixon et al., 2013). In the UK, one of the most recent extensive uses to QR code systems is the NHS COVID-19 application. Venues are instructed to download and print out NHS QR code posters so customers can scan

Figure 4. Pie charts showing awareness of and familiarity with Quick Response codes.



the displayed official NHS QR code when checking in. This helps track and trace recent close contacts to a positive coronavirus and advise others accordingly to self-isolate as necessary, thus helping to protect lives. In China, colour-coded QR codes have also been used as an electronic certificate to differentiate the health status of the individuals with regards to COVID-19 and present them when using public services. This digital approach was reported successful to contain the number of positive COVID-19 cases and tackle the pandemic (Nakamoto et al., 2020).

Patients' perceptions

Currently, the collective data have outlined the majority of the participants preferring the use of QR codes in delivering PILs

Figure 5. The modality of use to scan the Quick Response code.

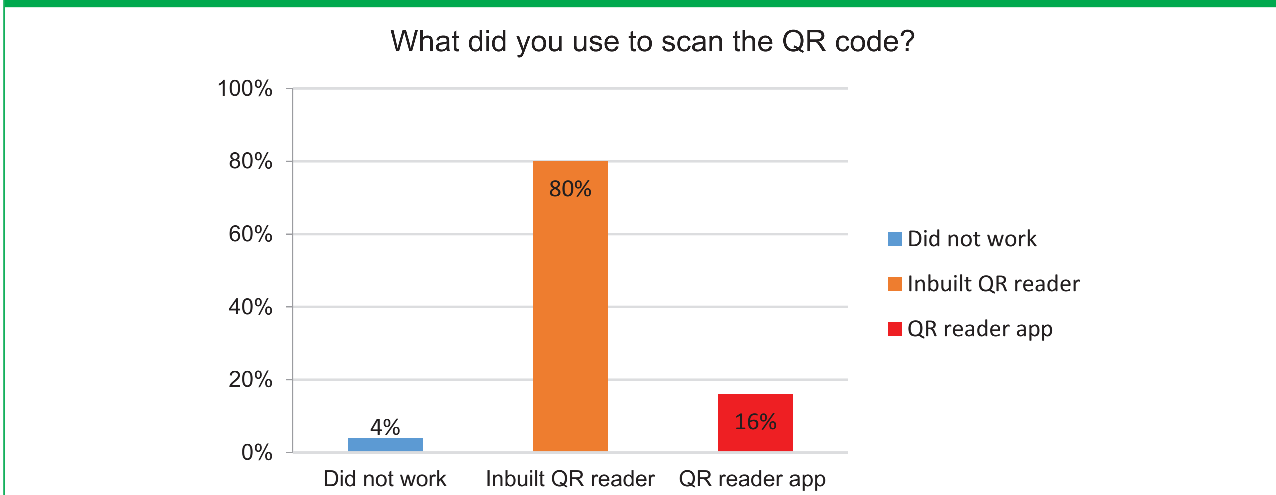


Figure 6. The ease with which the participants found scanning the Quick Response code.

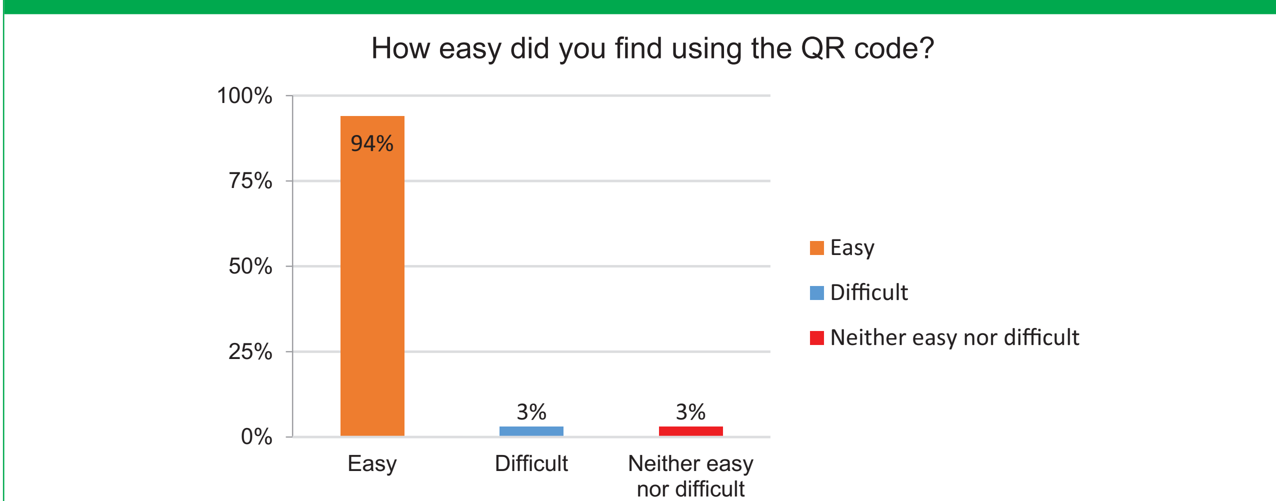


Figure 7. Respondents' preferences.

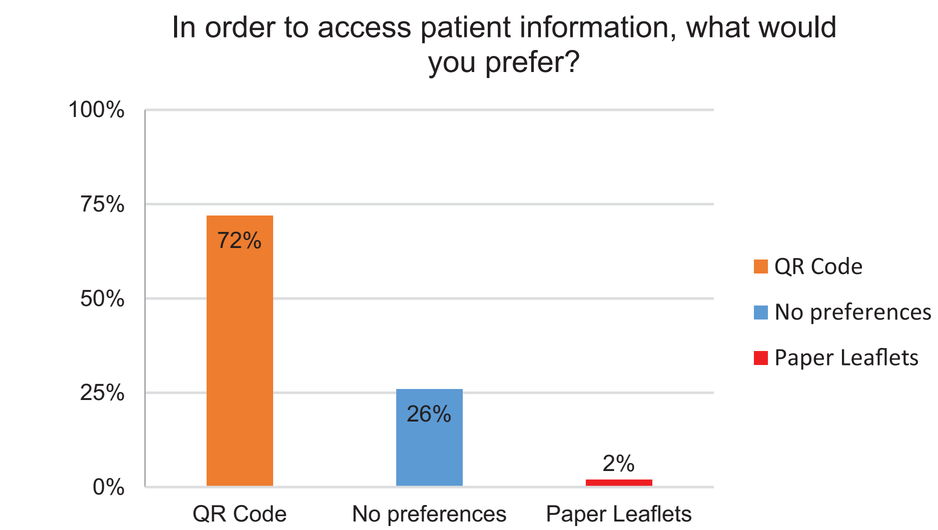
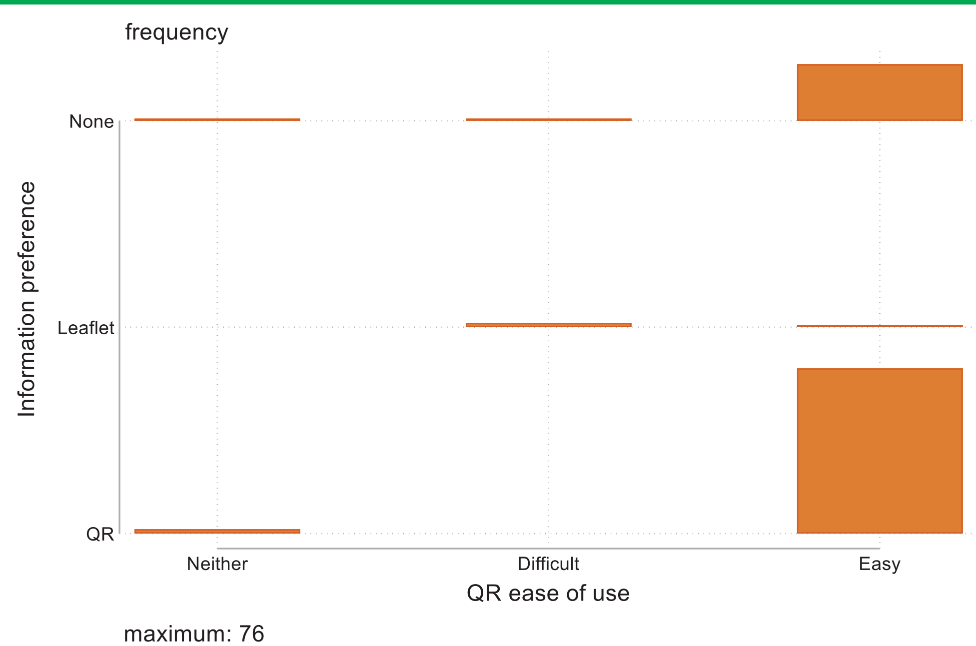


Figure 8. Graphical representation comparing the ease of use and participants' preferences.



(72%) compared to only 2% that reported paper leaflets and 26% being neutral. This is in line with previous evidence that showed satisfaction with the use of a QR code system, increasing participation, and an effectively linking technology to health information (Fawke et al., 2014; Bellot et al., 2015; Kafadarova et al., 2019; Kavitha and Sakthivel, 2019). The ease of use of QR codes showed statistically significant correlation with the participant preferences. Those who reported difficulty with the use of QR codes had not previously used QR codes and most of them preferred paper leaflets over the electronic version. Although patients can be more reluctant to change, with education and appropriate format of the health information leaflet to make it user friendly on the hospital website, increased acceptance could be anticipated (Hammar et al., 2016).

Effect of age and gender on patient preference

When looking at the participants demographics, although the attitude differed in relation to age and gender, this was not statistically significant. Adult participants showed higher preferences to the use of QR codes in dissemination of patient information compared to adolescents (76% and 67%, respectively), whereas 33% of adolescents and 20% of adults had no preference when choosing the method of information delivery. Only 4% of the adults and none of the adolescents favoured paper leaflets. With respect to gender, female participants reported preference to QR codes in 69%, 26% being neutral and 4% preferring paper leaflets compared to male participants who showed higher interest in QR codes (76%) and 24% being neutral. A recent study reported that age was a significant factor in determining the method of delivering patient information leaflets, with a more positive response towards an electronic version among those aged 55 years and younger compared to those aged 56 years and above; however, there were no significant differences between the sexes (Hammar et al., 2016).

Benefits of QR codes

In this study, QR codes demonstrated successful dissemination of data to patients. QR codes are instantly recognisable and universally readable by most smartphone cameras. The majority of patients in this study confirmed the familiarity and simplicity of using a QR code (94%). The touch-free capability of the QR code is the main advantage in the battle against COVID-19, obviating the need to exchange paper leaflets with patients thus improving infection control. In this study, QR codes were generated using a free website application, printed and laminated for the patient to scan, and therefore was almost free to generate and utilise, and supportive to the environment. Any information can also be easily updated on the website without the need of generating a new QR code. Traditional paper leaflets are not future-proof and would require reprinting and have a further impact on the environment. These advantages are confluent with the current literature where the use of QR code and smartphone technology were reported to improve knowledge, be more convenient, accurate, secure and environmentally friendly (Dixon et al., 2013; Uzun and Bilgin, 2016; Koh et al., 2017).

Barriers to QR codes

Nevertheless, QR codes are not without limitations. Drawbacks related to technicality, availability of smartphones, data security and patient perception of the unprofessional use of smartphones in a clinical setting have been documented (Bellot et al., 2015; Karia et al., 2019). For effective use, the information to which QR codes are linked to should be updated on the website. This will require time and personnel to ensure continuous provision of up-to-date information. Furthermore, after scanning the QR codes, patients were instructed to store the information either on a file on their smartphones or send it as an attachment to their email. Patients were advised they could log into the Trust website and retrieve the information leaflet should they not manage to locate the scanned information. During the initial phase of the COVID-19 pandemic, some adolescent patients attended the surgery alone without a parent and therefore a few did not have smartphone access and were excluded from taking part in this study.

Conclusion

With the recent disruptive period of COVID-19 and the abrupt changes in the NHS healthcare systems, QR codes can be a promising digital tool in delivering PILs. The majority of the patients found QR codes easy to use and preferred them over paper leaflets. QR codes have shown to be familiar, effective and a touch-free technology, therefore, improving infection control procedures with minimal patient contact with different surfaces and objects.

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